

Bench	Task		Trials	Raw	+Workflow	+Skills	Raw Tok	WF Tok	SK Tok	ΔP_{SK-Raw}	ρ_{SK-Raw}
Terminal-Bench-2											
TB2	overfull-hbox		5	80.0%	80.0%	20.0%	555K	389K	597K	-60.0%	+7.6%
TB2	model-extraction-relu-logits		5	60.0%	40.0%	0.0%	482K	217K	356K	-60.0%	-26.2%
TB2	crack-7z-hash		5	80.0%	0.0%	40.0%	986K	323K	3.0M	-40.0%	+208.0%
TB2	path-tracing-reverse		5	80.0%	0.0%	40.0%	4.0M	838K	1.4M	-40.0%	-64.0%
TB2	query-optimize		5	40.0%	0.0%	0.0%	642K	243K	894K	-40.0%	+39.3%
TB2	torch-tensor-parallelism		5	40.0%	20.0%	0.0%	174K	108K	191K	-40.0%	+9.7%
TB2	financial-document-processor		5	60.0%	60.0%	20.0%	1.1M	400K	434K	-40.0%	-60.5%
TB2	adaptive-rejection-sampler		5	20.0%	60.0%	0.0%	1.7M	374K	2.0M	-20.0%	+20.0%
TB2	tune-mjcf		5	20.0%	0.0%	0.0%	453K	237K	0K	-20.0%	-100.0%
TB2	chess-best-move		5	80.0%	60.0%	80.0%	361K	255K	379K	+0.0%	+5.1%
TB2	hf-model-inference		5	60.0%	40.0%	60.0%	1.1M	381K	784K	+0.0%	-30.0%
TB2	mteb-leaderboard		5	60.0%	0.0%	60.0%	19.8M	1.0M	3.5M	+0.0%	-82.5%
TB2	path-tracing		5	80.0%	0.0%	80.0%	2.2M	353K	1.3M	+0.0%	-37.7%
TB2	build-pov-ray		5	80.0%	60.0%	100.0%	1.4M	978K	857K	+20.0%	-37.1%
TB2	fix-code-vulnerability		5	80.0%	80.0%	100.0%	1.4M	893K	825K	+20.0%	-41.5%
TB2	large-scale-text-editing		5	80.0%	40.0%	100.0%	403K	304K	239K	+20.0%	-40.7%
TB2	log-summary-date-ranges		5	80.0%	100.0%	100.0%	163K	126K	135K	+20.0%	-17.2%
TB2	mcmc-sampling-stan		5	80.0%	0.0%	100.0%	10.8M	1.0M	9.7M	+20.0%	-9.8%
TB2	portfolio-optimization		5	80.0%	100.0%	100.0%	391K	285K	306K	+20.0%	-21.6%
TB2	regex-chess		5	80.0%	0.0%	100.0%	3.0M	394K	2.1M	+20.0%	-29.9%
TB2	rstan-to-pystan		5	80.0%	0.0%	100.0%	4.1M	877K	2.6M	+20.0%	-37.0%
TB2	sanitize-git-repo		5	80.0%	60.0%	100.0%	2.2M	940K	835K	+20.0%	-62.4%
TB2	winning-avg-corewars		5	80.0%	40.0%	100.0%	4.7M	409K	3.1M	+20.0%	-32.8%
TB2	pypi-server		5	20.0%	40.0%	40.0%	499K	378K	404K	+20.0%	-19.1%
TB2	pytorch-model-recovery		5	40.0%	80.0%	80.0%	794K	242K	941K	+40.0%	+18.4%
TB2	write-compressor		5	60.0%	0.0%	100.0%	248K	56K	423K	+40.0%	+70.5%
TB2	break-filter-js-from-html		5	40.0%	0.0%	100.0%	376K	49K	750K	+60.0%	+99.3%
TB2	cancel-async-tasks		5	40.0%	0.0%	100.0%	69K	69K	138K	+60.0%	+99.4%
TB2	extract-elf		5	40.0%	80.0%	100.0%	722K	242K	246K	+60.0%	-65.9%
TB2	dna-insert		5	20.0%	60.0%	100.0%	454K	335K	355K	+80.0%	-21.7%
TB2	qemu-startup		5	20.0%	40.0%	100.0%	758K	355K	229K	+80.0%	-69.8%
SkillsBench											
SB	paper-anonymizer		5	40.0%	20.0%	0.0%	940K	874K	756K	-40.0%	-19.6%
SB	weighted-gdp-calc		5	80.0%	60.0%	40.0%	633K	415K	549K	-40.0%	-13.3%
SB	grid-dispatch-operator		5	40.0%	60.0%	20.0%	635K	677K	715K	-20.0%	+12.7%
SB	react-performance-debugging		5	40.0%	60.0%	20.0%	3.8M	5.4M	5.2M	-20.0%	+35.0%
SB	energy-market-pricing		5	60.0%	20.0%	40.0%	866K	1.3M	971K	-20.0%	+12.1%
SB	invoice-fraud-detection		5	60.0%	20.0%	40.0%	359K	317K	430K	-20.0%	+19.7%
SB	protein-expression-analysis		5	80.0%	80.0%	80.0%	529K	460K	396K	+0.0%	-25.1%
SB	dapt-intrusion-detection		5	80.0%	100.0%	100.0%	277K	259K	358K	+20.0%	+29.2%
SB	earthquake-phase-association		5	80.0%	60.0%	100.0%	2.1M	2.4M	907K	+20.0%	-57.8%
SB	glm-lake-mendota		5	80.0%	80.0%	100.0%	2.0M	1.9M	504K	+20.0%	-75.0%
SB	hvac-control		5	80.0%	100.0%	100.0%	220K	401K	329K	+20.0%	+49.7%
SB	virtualhome-agent-planning		5	40.0%	100.0%	60.0%	284K	206K	173K	+20.0%	-39.1%
SB	data-to-d3		5	60.0%	60.0%	80.0%	882K	642K	654K	+20.0%	-25.9%
SB	fix-erlang-ssh-cve		5	60.0%	100.0%	80.0%	2.3M	1.4M	1.0M	+20.0%	-54.4%
SB	r2r-mpc-control		5	60.0%	40.0%	80.0%	526K	544K	820K	+20.0%	+55.9%
SB	adaptive-cruise-control		5	20.0%	20.0%	60.0%	5.7M	1.9M	3.5M	+40.0%	-39.8%
SB	fix-druid-loophole-cve		5	20.0%	20.0%	60.0%	13.4M	10.1M	12.5M	+40.0%	-6.5%
SB	fix-visual-stability		5	60.0%	100.0%	100.0%	4.9M	4.2M	6.4M	+40.0%	+30.2%
SB	dialogue-parser		5	40.0%	60.0%	100.0%	488K	427K	551K	+60.0%	+13.0%
SB	financial-modeling-qa		5	40.0%	40.0%	100.0%	827K	667K	435K	+60.0%	-47.4%
SB	video-filler-word-remover		5	40.0%	100.0%	100.0%	2.8M	1.5M	793K	+60.0%	-71.4%
SB	gravitational-wave-detection		5	20.0%	100.0%	80.0%	2.5M	2.3M	411K	+60.0%	-83.8%
SB	organize-messy-files		5	20.0%	40.0%	80.0%	602K	420K	806K	+60.0%	+33.7%
SB	trend-anomaly-causal-inference		5	20.0%	20.0%	80.0%	1.9M	3.8M	3.2M	+60.0%	+66.6%
Terminal-Bench-Pro											
TB-Pro	implement-nonogram-puzzle-solver		5	60.0%	60.0%	0.0%	130K	315K	126K	-60.0%	-3.0%
TB-Pro	python-jacobi-eigen-solver-conversion		5	60.0%	0.0%	0.0%	198K	178K	236K	-60.0%	+19.4%
TB-Pro	fix-gomoku-win-detection		5	80.0%	100.0%	40.0%	92K	89K	142K	-40.0%	+54.4%
TB-Pro	train-sarsa-taxi-agent		5	60.0%	100.0%	20.0%	422K	1.1M	320K	-40.0%	-24.0%
TB-Pro	portfolio-risk-analyzer-script		5	20.0%	0.0%	0.0%	182K	205K	191K	-20.0%	+4.8%
TB-Pro	train-fraud-detection-model		5	40.0%	20.0%	20.0%	180K	125K	162K	-20.0%	-10.3%
TB-Pro	advanced-json-to-rfc4180-csv-converter		5	60.0%	60.0%	60.0%	401K	397K	476K	+0.0%	+18.8%
TB-Pro	build-uci-chess-engine-to-beat-multiple-bots		5	40.0%	20.0%	40.0%	1.1M	1.0M	438K	+0.0%	-60.5%
TB-Pro	decode-go-ctf-credentials		5	80.0%	100.0%	80.0%	958K	580K	438K	+0.0%	-54.3%
TB-Pro	fix-xor-neural-network-instability		5	80.0%	20.0%	80.0%	544K	459K	307K	+0.0%	-43.5%
TB-Pro	implement-chemical-equilibrium-solver		5	20.0%	0.0%	20.0%	215K	499K	373K	+0.0%	+73.3%
TB-Pro	implement-depgraph-dependency-resolver		5	20.0%	60.0%	20.0%	468K	310K	336K	+0.0%	-28.2%
TB-Pro	optimize-urdf-robot-for-pybullet		5	60.0%	80.0%	60.0%	1.6M	1.9M	1.8M	+0.0%	+12.3%
TB-Pro	reverse-engineer-kvstore-binary-protocol		5	80.0%	60.0%	80.0%	810K	380K	838K	+0.0%	+3.4%
TB-Pro	train-matrix-factorization-embeddings		5	60.0%	100.0%	60.0%	351K	188K	268K	+0.0%	-23.8%
TB-Pro	validate-and-solve-sudoku		5	60.0%	80.0%	60.0%	438K	184K	245K	+0.0%	-44.1%
TB-Pro	improve-code-similarity-feature-extraction		5	80.0%	80.0%	100.0%	803K	471K	557K	+20.0%	-30.7%
TB-Pro	normalize-invoice-pdfs-to-csv		5	40.0%	0.0%	60.0%	212K	170K	220K	+20.0%	+3.5%
TB-Pro	secure-postgres-schema-migration		5	80.0%	80.0%	100.0%	634K	517K	590K	+20.0%	-6.9%
TB-Pro	train-loan-default-logreg-model		5	80.0%	100.0%	100.0%	147K	101K	142K	+20.0%	-3.5%
TB-Pro	flood-escape-path-planner		5	60.0%	80.0%	80.0%	214K	175K	210K	+20.0%	-1.8%
TB-Pro	present-known-plaintext-key-recovery		5	60.0%	100.0%	80.0%	5.9M	2.5M	1.9M	+20.0%	-68.2%
TB-Pro	repair-broken-shell-data-pipeline		5	60.0%	100.0%	80.0%	759K	842K	539K	+20.0%	-28.9%
TB-Pro	recover-and-sanitize-git-db-credential		5	20.0%	0.0%	60.0%	548K	368K	458K	+40.0%	-16.6%
TB-Pro	compare-cmake-build-types-performance		5	60.0%	100.0%	100.0%	111K	111K	132K	+40.0%	+19.5%
TB-Pro	mongodb-sales-aggregation-engine		5	60.0%	80.0%	100.0%	167K	112K	148K	+40.0%	-11.5%
TB-Pro	setup-mlflow-sqlite-server		5	60.0%	60.0%	100.0%	1.0M	899K	224K	+40.0%	-78.3%
TB-Pro	text-image-ocr-pipeline		5	60.0%	100.0%	100.0%	109K	93K	129K	+40.0%	+18.4%
TB-Pro	implement-tensor-parallel-matmul		5	40.0%	100.0%	100.0%	165K	196K	196K	+60.0%	+18.7%
TB-Pro	jq-github-contributor-report		5	40.0%	80.0%	100.0%	111K	175K	285K	+60.0%	+157.9%
TB-Pro	python-sql-injection-sanitizer		5	40.0%	100.0%	100.0%	201K	164K	171K	+60.0%	-14.9%
TB-Pro	regex-bitcoin-p2pkh-extraction		5	40.0%	80.0%	100.0%	65K	44K	96K	+60.0%	+47.5%
TB-Pro	sokoban-bfs-move-with-fallback		5	20.0%	100.0%	100.0%	111K	141K	179K	+80.0%	+61.0%